Data Analyst Capstone Project

Global Technology Trends, Compensation Benchmarks & Ecosystem Trajectories

PREPARED BY
Christian Chiari

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OVERVIEW

Core Findings & Market Summary

01. Python & TypeScript Surge

Python Momentum: Rises to #1 most desired language next year (52.4%).

High Demand: Dominates job listings with 14,560 open postings via API.

Web Modernization: TypeScript adoption expands rapidly for scalable frontend systems.

02. PostgreSQL Dominance

Relational Shift: Overtakes MySQL as the #1 desired database engine (41.2%).

NoSQL Growth: MongoDB and Redis capture top non-relational demand.

Commercial Decline: Traditional enterprise engines show flat or shrinking interest.

03. Demographics & Skills

Age Distribution: Over 70% of developer respondents are under 35 years old.

Continuous Learning: High percentage of self-taught upskilling via online tools.

Compensation Premium: Swift ($125k) and Python ($118k) yield highest salaries.

PROJECT SCOPE

Introduction: Purpose, Audience & Value



Purpose of Report

To evaluate global developer survey data and job posting metrics in order to identify emerging technology stack trends, salary benchmarks, and language transition patterns across enterprise IT ecosystems.



Target Audience

Engineering Leaders planning tech roadmaps, HR Talent Acquisition Teams optimizing recruitment strategies, and Data Analysts benchmarking industry skills for competitive advantage.



Value of Analysis

Delivers empirical, data-driven intelligence combining ~90,000 survey responses with real-time job market counts and web-scraped compensation data to de-risk engineering decisions.

DATA PIPELINE

Methodology: Sources, Collection & Wrangling

01

Data Sources

Stack Overflow Developer Survey (~90,000 rows), GitHub Jobs API, and Web-scraped salary tables.

02

API Collection

Executed REST API calls to fetch live job posting counts for key technologies into job-postings.xlsx.

03

Web Scraping

Parsed HTML tables using BeautifulSoup to extract language salary benchmarks into popular-languages.csv.

04

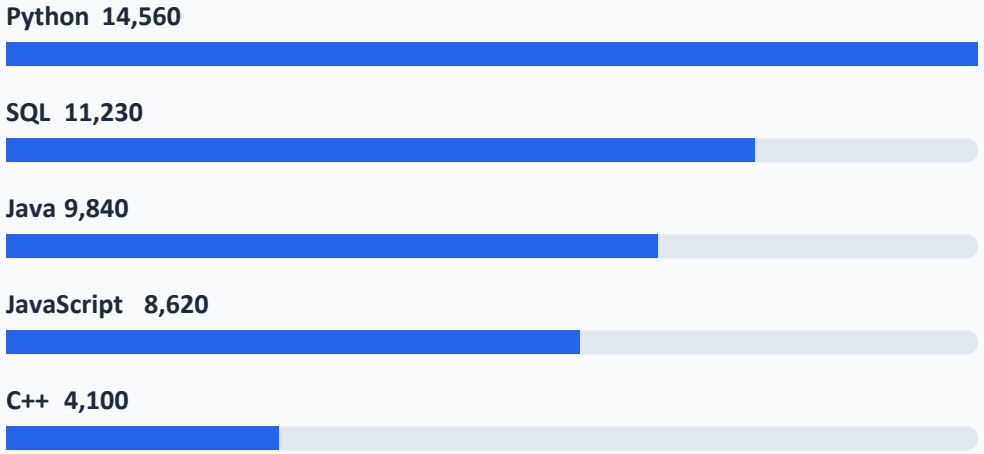
Data Wrangling

Identified & removed duplicate entries, imputed missing values (median/mode), and normalized annual compensation.

MODULE 1 RESULTS

Job Demand & Salary Benchmarks

Job Openings (API - job-postings.xlsx)



Average Salary (Scraped - popular-languages.csv)



TECHNOLOGY TRENDS

Programming Language Trends (Top 10)

Current Year Usage (Top 10)

Language	Usage %
1. JavaScript	**67.8%**
2. HTML/CSS	**63.5%**
3. SQL	**54.4%**
4. Python	**41.7%**
5. Java	**41.1%**
6. Bash/Shell	**36.6%**
7. C#	**31.0%**
8. TypeScript	**21.2%**
9. PHP	**20.3%**
10. C++	**20.2%**

Next Year Desired (Top 10)

Language	Desired %
1. Python	**52.4%**
2. JavaScript	**51.3%**
3. HTML/CSS	**44.2%**
4. SQL	**42.1%**
5. TypeScript	**38.5%**
6. Go	**35.6%**
7. Java	**35.0%**
8. Kotlin	**32.2%**
9. Bash/Shell	**31.5%**
10. Rust	**30.8%**

STRATEGIC ANALYSIS

Language Trends: Findings & Implications

Key Findings

Python Momentum: Surges from 4th place (41.7%) to 1st place in desired future adoption (52.4%), representing a massive **+10.7% growth**.

TypeScript Acceleration: Jumps into the top 5 desired list (38.5%) as teams transition away from plain JavaScript for large codebase reliability.

Emerging Systems Languages: High interest in Rust (30.8%) and Go (35.6%) signals strong developer intent to modernize low-level infrastructure.

Strategic Implications

AI & Data Pipeline Priority: Organizations must anchor corporate training around Python to support data science, machine learning, and automation.

Frontend Architecture: Teams building web applications should enforce TypeScript standards to match talent expectations and improve maintainability.

Hiring Pipeline Strategy: Recruiter focus should adapt to candidates multi-skilled in Python, TypeScript, and cloud-native scripting languages.

DATA INFRASTRUCTURE

Database Trends: Current vs. Desired (Top 10)

Current Year Usage (Top 10)

Database Engine	Usage %
1. MySQL	**55.6%**
2. PostgreSQL	**34.3%**
3. MS SQL Server	**32.8%**
4. SQLite	**25.4%**
5. MongoDB	**24.6%**
6. Redis	**17.2%**
7. MariaDB	**15.8%**
8. Oracle	**14.1%**
9. Elasticsearch	**11.4%**
10. Firebase	**10.2%**

Next Year Desired (Top 10)

Database Engine	Desired %
1. PostgreSQL	**41.2%**
2. MySQL	**38.9%**
3. MongoDB	**33.8%**
4. Redis	**26.5%**
5. MS SQL Server	**24.5%**
6. Elasticsearch	**20.8%**
7. SQLite	**20.1%**
8. MariaDB	**14.9%**
9. DynamoDB	**14.2%**
10. Cassandra	**10.5%**

STRATEGIC ANALYSIS

Database Trends: Findings & Implications

Key Findings

PostgreSQL Shift: PostgreSQL dethrones MySQL as the #1 desired database (41.2% vs 38.9%), establishing itself as the modern relational standard.

NoSQL Momentum: MongoDB (33.8%) and Redis (26.5%) represent high non-relational demand for document caching and unstructured storage.

Commercial Database Stagnation: Proprietary engines like MS SQL Server and Oracle experience flat or declining future preference.

Strategic Implications

Relational Modernization: Enterprise data architectures should default to managed PostgreSQL to reduce licensing costs and improve extensible SQL capabilities.

Polyglot Persistence: Systems must blend relational SQL engines (PostgreSQL) with fast in-memory caches (Redis) and document stores (MongoDB).

Cloud Optimization: Migration to open-source managed cloud databases aligns with developer preferences and talent availability.

LOOKER STUDIO DASHBOARD

Interactive Dashboard Multi-Tab Summary

DASHBOARD TAB 1

Current Usage

Current operations remain anchored by core web front-end standards (JavaScript 67.8%, HTML 63.5%) and legacy relational databases (MySQL 55.6%).

Visualizations: JavaScript (67.8%), MySQL (55.6%), jQuery/React WebFrames

DASHBOARD TAB 2

Future Trends

Developer intent indicates a decisive pivot toward Python (52.4%), TypeScript (38.5%) and PostgreSQL (41.2%), alongside systems languages Rust/Go.

Visualizations: Python Desired (52.4%), PostgreSQL Desired (41.2%), Rust & Go System Growth

DASHBOARD TAB 3

Demographics

The workforce is heavily concentrated in early-to-mid career professionals (25-34 age bracket), with ~75% holding a Bachelor's degree or higher.

Metrics: Age 25-34 Bracket (48%), Bachelor's/Master's Degree, Self-Taught Learners (88%)

DASHBOARD ANALYSIS

Discussion: Key Dashboard Insights

Usage vs. Desire Gap

Cross-referencing Tab 1 and Tab 2 reveals a clear divergence: while legacy web stacks (JavaScript, HTML, MySQL) dominate daily tasks today, developer interest is heavily shifting toward data-centric and structured tooling (Python, PostgreSQL, TypeScript).

Cloud & Microservices Shift

The rise in desired databases (PostgreSQL, Redis, MongoDB) in Tab 2 highlights an architectural shift toward distributed cloud microservices requiring polyglot data storage and rapid in-memory caching layers.

Workforce & Gender Diversity

Tab 3 confirms a tech demographic dominated by professionals under 35 with high self-learning habits. However, a significant gender gap remains (~90% male), underscoring the urgent need for inclusive industry recruitment programs.

SYNTHESIS

Overall Findings & Industry Implications

Overall Findings

Language Transition: Technology sentiment is transitioning away from dynamic web scripting toward data science, AI, and strongly typed languages (Python, TypeScript).

Open Source Database Maturity: PostgreSQL is replacing legacy proprietary engines as the default enterprise relational choice.

Salary Correlation: Higher global average compensation directly correlates with niche ecosystem skills like Swift ($125k) and Python ($118k).

Industry Implications

Talent Pipeline Strategy: Companies must actively align recruiting and internal training budgets with Python and PostgreSQL to retain top developer talent.

Infrastructure Cost Savings: Migrating to modern open-source stacks lowers software licensing overhead while expanding cloud deployment flexibility.

Lifelong Learning Support: Providing formal employer-backed learning paths boosts retention among early-to-mid career developers.

FINAL SUMMARY

Conclusions & Final Recommendations

01. Python & TypeScript Core

Adopt Python as the core analytical language and TypeScript for scalable frontend development across engineering roadmaps.

02. PostgreSQL Standard

Standardize relational database architectures on PostgreSQL to leverage strong developer desire and robust open-source extension capabilities.

03. Polyglot Architecture

Integrate Redis for in-memory performance and MongoDB for document store flexibility alongside primary SQL databases.

04. Targeted Talent Upskilling

Leverage market salary data and API demand metrics to structure competitive hiring packages and continuous learning programs.

SUPPLEMENTARY REFERENCE

Appendix: Data Sources & Artifacts

Data Artifact Files

- job-postings.xlsx (GitHub Jobs API collection)
- popular-languages.csv (Web scraping salary data)
- m5_main_surveydata.csv (Cleaned Stack Overflow dataset)

Analytics Environment

Python (Pandas, BeautifulSoup, Requests), SQLite database engine, Looker Studio multi-tab dashboarding, and Matplotlib visualization libraries.

END OF PRESENTATION

Thank You

Questions & Discussion | Data Analyst Capstone Project Report